



Aim

Investigate the practical tasks and required equipment needed for making wine.

INTRODUCTION

The following is an outline of what is involved in making wine using grapes.

Stage 1 - Harvesting

Grapes are picked and brought to the press house as soon as possible after picking. The grapes may be handpicked or harvested using machinery. The timing of harvesting is an important step, although it may be dictated by weather conditions. Grapes are tasted prior to picking to assess the balance of sweetness and acidity because this will influence the taste of the wine. If they are deemed ready, the grapes are picked and this flavour is locked in.

Stage 2 - Crushing and Pressing

Any bruised or inferior grape bunches are separated out from the rest. Those considered good enough are fed into a crushing machine to extract the juice. The crusher replaces the need to stomp on the grapes and also removes stems, a task which used to be done by hand. If they are not removed, stems will increase the pH of the juice which, in turn, decreases colour intensity and fruitiness.

Crushing is the process of mechanically breaking the grape skins in order to expose the juice to the yeast and permit alcoholic fermentation. Tannins will be extracted mainly from the grape skins however some stems may be left in or even added in small amounts to increase tannin levels, especially in red wines.

As already mentioned, white wine grapes have their skin removed. This is to remove tannins as well as unwanted colour. White wine grapes are removed from their skins as soon as possible after picking.

On the other hand, red grapes being used to make red wine have their skins left in during fermentation (if skins are removed red grapes would produce white wine). Red wine is left in contact with the skins to acquire flavour and tannins. 'Maceration' is the name of the process whereby crushed grapes are allowed to soak in their juice, for the purpose of extracting phenols from the solids. Red wines obtain their distinctive 'structure' during this phase. They develop colour, flavour and texture. Full-bodied reds, such as Cabernet Sauvignons may be left to macerate for up to 21 days. With rosé, skins from red grapes are removed a short time after fermentation starts.

Pressing is the process of juice extraction from the crushed grapes by pressure. Pressed white grapes produce a cloudy juice that will require clarification during the final stages of vinification. The juice derived from pressing is usually fermented separately.



Stage 3 - Alcoholic Fermentation

This stage is where the juice, or must, is converted into wine and hence is an important step in good winemaking. Although the juice will begin to ferment naturally within hours due to the action of airborne yeasts, winemakers generally add cultured yeasts to ensure a consistent and more predictable end-product. Yeasts convert sugar into alcohol during fermentation, so the strain of yeast used is vital. As fermentation progresses, all the sugar is converted into alcohol. The winemaker needs to control factors such as sugar concentration, acidity and pH, temperature, and sulphur dioxide (SO₂) levels. If a sweet wine is required, then fermentation may be stopped before all the sugar is converted to alcohol.

Fermentation should be monitored regularly by assessing the changing sugar concentration (Brix) in the container. Typically, this is done using a refractometer which is an instrument used to measure the angle of refraction through the juice and thereby give an indication of the amount of sugar. The instrument is calibrated using Brix units where 1 Brix = 1g of sugar per 100mg liquid i.e. 1% sugar concentration.

“Chaptalisation” is a process that involves adding sugar to the wine “must”. It is used in situations where the grapes have low levels of natural sugars, such as in a cold climate or a cooler year. The process may involve adding cane sugar directly into the wine or dissolving sugar in a small sample of must; then adding that sugary must. It can however be problematic if some sugar does not dissolve properly. In some colder wine growing areas, liquid sugar solutions are produced and sold to wine makers for chaptalisation use.

Sugar Free Extract (also referred to as extract) refers to a minerals and nonvolatile chemicals dissolved in wine. Lower extract levels usually result from using unripe grapes and higher extract levels from using riper grapes. Extract levels in wine generally vary from 17 to 30gm per litre. Higher extract levels are associated with a wine being judged to have more “body”.

Secondary, or malolactic, fermentation is often used after alcoholic fermentation. It is more common for red wine production and for some whites like chardonnay. During this process, malic acid which has a tart taste is converted into lactic acid which has a smoother taste. Whilst once again malolactic fermentation can occur naturally it is usually augmented by the addition of bacteria. Malolactic fermentation can also be stopped if it is not wanted, for example when making fruity wines.

Stage 4 - Clarification

Clarification removes cloudiness and haze from wine. A good wine should be free of suspended particles which interfere with clarity. The particles removed include proteins, tannins and dead yeast cells. Sedimentation and racking are traditional clarification methods. In commercial production, the wine is racked into another container such as a wine barrel. Clarification then proceeds to fining or filtration. Fining involves adding fining agents to clarify the wine. The agent may draw unwanted particles to it which can then be removed (e.g. clay may be used to do this if it is placed on the bottom of the barrel. The wine can then be racked into another barrel. Filtration involves removing larger particles by trapping them in a filter. Once again, the clarified wine can be racked into another barrel.

Sometimes wine from two or more varieties of grape is blended to yield the final drinkable product. Blending has an extensive tradition and is responsible for the distinct flavour and characteristics of regional wines. This process is also called assemblage. Blending between vintages is used to ensure a consistent, good quality product. Blending allows the winemaker to instill a good balance to the wine they produce. If done properly, blending establishes a good balance of complexity of smell (bouquet) and taste, body, fruitiness, colour intensity, acidity, oak character, and many other important characteristics.

Stage 5 - Aging and Bottling

The wine can be bottled immediately after clarification or it can be left to age in other vessels. Aging can enhance the characteristics the winemaker has infused into the wine. Complexity of flavour and feel can be improved through oak barrelling, but wines can also be left to age in bottles or stainless-steel tanks. The main difference between storage in steel or glass containers versus barrels is the very slow diffusion of air and evaporation that occurs through barrels. Some wines are actually fermented in barrels, resulting in complex flavours.

Bottling is the final stage in the winemaking process whether the wine is intended for distribution and immediate consumption or for further ageing. The main goal of bottling is to prevent the wine from further exposure to air. The actual shape of the bottle can easily identify wines of particular types e.g. Burgundy, Champagne, or Port. Wines to be aged are often stored in green or brown-tinted bottles. Traditionally premium wines were sealed with a natural cork, however, these days there are alternatives. Synthetic corks are widely available and screw caps that offer effective sealing are now commonly used. However, a wine which is corked

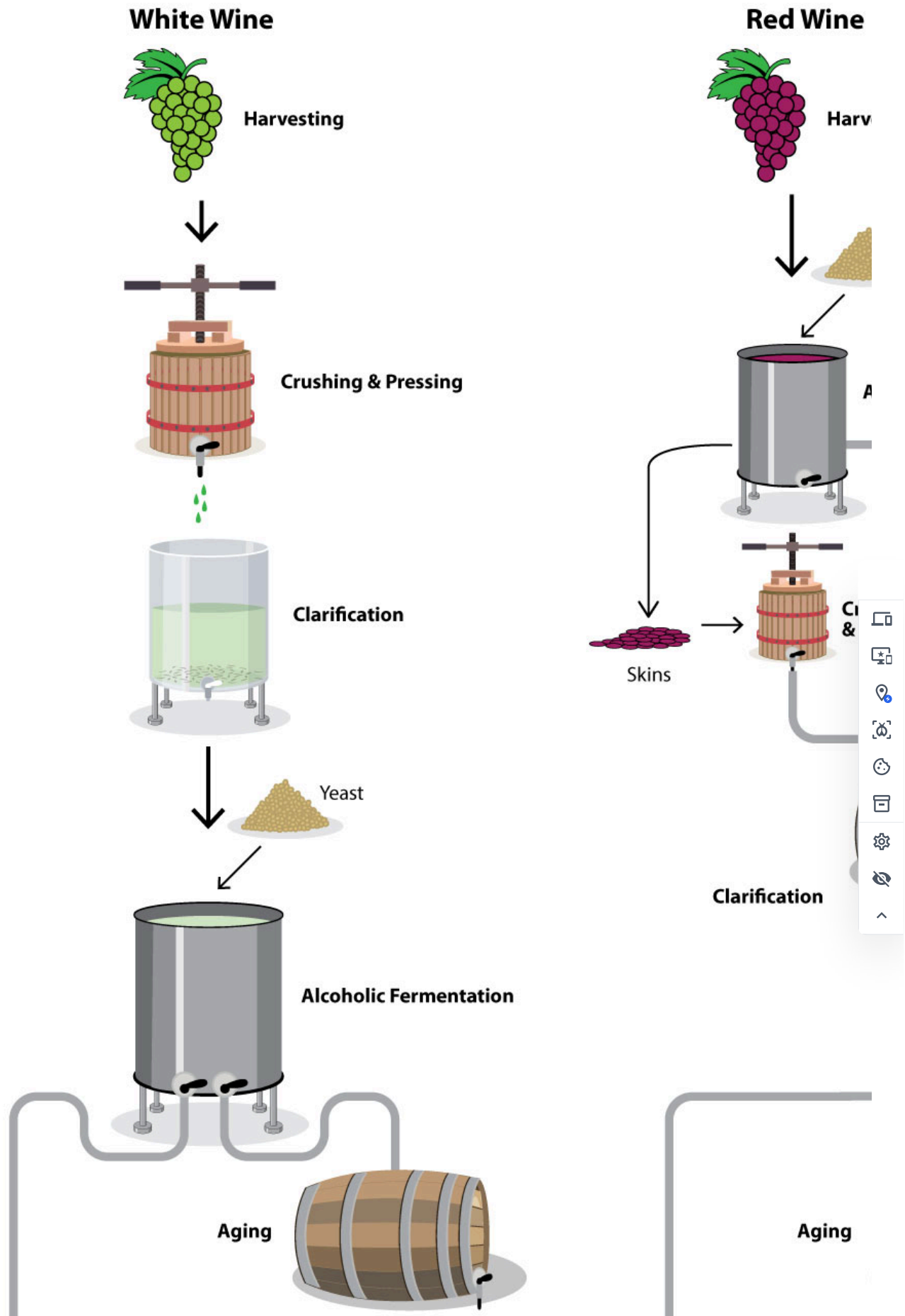


using natural cork will be able to age better in the bottle because a tiny amount of air is able to pass through the cork. Capsules are often placed over dry corked necks to retard mould formation and improve the look of the bottle. Plastic types are often called shrink caps because they have to be heat shrunk onto the neck. Metal types are also used.

Cellaring may be used with some wines which can be enhanced by time spent in storage. Red wines rich in tannins may taste better after several years or ten to fifteen years. Lower alcohol wines, and reds with less tannins, are generally not meant to be stored. However, even when wines are stored the storage conditions will alter the effect of the wine's aging process. Cellaring conditions should provide a constant temperature between 10°C and 13°C, constant humidity between 55 and 75%, complete darkness, and a clean atmosphere, e.g. no strong smells. The activities of many different types of yeasts and bacteria needs to be controlled to ensure the final result is what you intended.



WINE MAKING PROCESS



Bottle
oaked
WineBottle
Unoaked
Wine

SULPHITING

Sulphiting is the process of adding sulphur dioxide to wine. This is commonly done by adding sodium or potassium metabisulphite. 200mg added to 1 litre of wine, will result in 100ppm of sulphur dioxide. Sulphurous acid is also sometimes used.

Timing of sulphiting is critical. It is commonly added when grapes are crushed or when the “must” is being prepared. If added during fermentation it can cause development of chemicals that negatively affect the wine flavour.

CLARIFICATION AND STABILISATION

As wine develops, sediment settles on the bottom of the tank, container, or bottle. As the fermentation process slows and all but stops, most yeasts also settle on the bottom. This sediment is called the “lees”. At this point in time the wine has no remaining “free” sulphur dioxide.

Provided malo-lactic fermentation is not required, carefully determined levels of sulphur dioxide are added at this point in order to deter oxidation and bacterial contamination; and hasten clarification and settling of sediment. At this stage:

- Dry wines need 20 to 30ppm of free sulphur dioxide
- Wines with 2% sugars need 30 to 40ppm free sulphur dioxide
- Sweeter wines with 5% sugars need 50ppm of free sulphur dioxide

Sulphur dioxide levels in young wines should be checked fortnightly

Most solid particles are able to be removed from wine through settling to form the lees (sediment on the bottom). Some fine particles, in particular proteins, do, however, remain suspended, reducing clarity in the wine.

Wine is clarified by 3 processes – racking, fining and filtering.

Racking

This involves siphoning the wine from the lees or sediment. It should be done soon after fermentation finishes. If wine is left with the lees for too long, the lees can significantly change the flavour of the wine as yeast cells decompose in the lees. Racking may need to be done multiple times. Sulphur dioxide levels should be monitored and adjusted as it is carried out.

Fining

Finings are materials that when added to wine will cause suspended proteins to stick together, become heavier, and fall to the bottom of the wine. Commonly used finings are bentonite and gelatine.

- Bentonite is added at 0.5 gm per litre of wine
- Gelatine is added at 0.06 gm per litre of wine

Filtering

Commercially available wine filters will remove most suspended particles; but generally do not remove all microorganisms. Wines that contain sugars can contain residual microorganisms, and to remove those, requires a special type of filter called a “sterile filter”.



If a wine is filtered with a sterile filter, it is very important to avoid exposing the wine to contamination afterwards. Bottles, corks and equipment used after sterile filtering must be sterilised and the environment and people who come in contact with sterile filtered wine need to maintain perfect hygiene until the wine is in a sealed container.

PRESERVATIVES (OR NOT)

Potassium sorbate may be dissolved in water and added to wine to create sorbic acid, which will act as a preservative, stabilising the wine and preventing microorganisms from becoming active and changing it. This is more critical with sweeter wines, which need more preservative than dry wines.

Sulphur dioxide is the main preservative used throughout the wine-making process. It can be labelled as 'preservative 220 added.' At 450+ppm, it can provide similar protection to sorbic acid; however, at that level, sulphur dioxide can negatively affect flavour. The maximum amount of sulphur that can be added in Australian wine is 300 mg/L.

Preservative-free wines are increasingly in demand. But the term 'preservative-free' is misleading. A small amount of sulphur dioxide is naturally released by the grapes during fermentation. All wine contains trace amounts of naturally produced preservatives. Wines produce their own natural preservatives: alcohol, tannin (phenolics), and natural acidity. Red wines have more of these natural preservatives.

Getting the highest-quality grapes in pristine condition ensures it is possible to add a lot less sulphur dioxide. Regardless of the quality, the real issue with wines containing sulphites is the problems they cause for people with allergies—headaches, sneezing, and others.

Hence the wines labelled preservative free means, the winemaker has not added any preservatives during the winemaking process.

HOW TO DETERMINE SUGAR LEVELS

Sugar is an essential component of wine production. The amount of sugar fermented determines the wine's alcohol level and the amount of residual sugar left in the wine. The concentration is usually expressed in Brix value (symbol °Bx) which is the percentage of sugar by weight in a liquid.

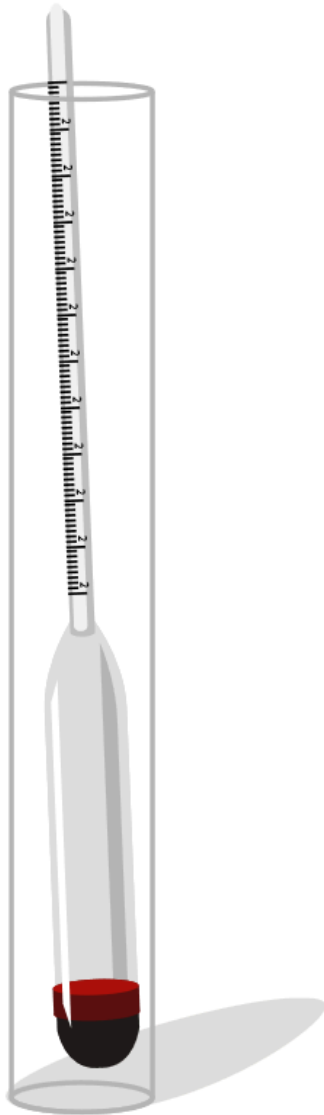
Sugar level above 2.2° Brix is measured using one of two instruments:

Refractometer:

The handheld refractometer is only used for field sampling before fermentation. Several berries are picked up from different parts of the vine and bunch are selected and crushed, a large drop of juice is then placed on the refractometer glass. After closing the lid, the refractive effects of sugar on the light will give a Brix reading on a scale. Temperature can also affect the reading and correction can be made as for hydrometer readings.

Hydrometer:





It is the preferred tool for measuring Brix reading any time after fermentation has begun. A

hydrometer is made of glass with weight at the bottom and typically has three scales of measurement Brix, Specific gravity, and Potential alcohol.

Procedure for using a hydrometer

1. Transfer about 100-200 ml of the sample into a 250 ml measuring cylinder
2. Ensure the hydrometer is dry and clean, then gently place the hydrometer in the cylinder and spin it by hand (to remove adhering air bubbles). The hydrometer will slowly rise if it is in the appropriate range.
3. The hydrometer must float freely without touching the sides and bottom of the cylinder. Once it settles, read the indicated value just below the liquid level.
4. Insert a thermometer into the juice. Check if the temperature corresponds to the temperature at which the hydrometer is calibrated. If it is outside the range shown in table 1, cool or warm the sample within the range and repeat the measuring procedure.
5. Thus, if the temperature of the sample is 16 °C and the Brix reading is 8%, the corrected value is $8 - 0.19 = 7.81$, if the temperature of the sample is 24°C and the Brix reading is 20%, then the corrected value is $20 - 0.26 = 19.74$.

For corrected Brix value search online as:

Temperature corrections to readings of Brix hydrometers (calibrated at 20 °C) or

Correction table for hydrometer reading taken at various temperature for 20 °C Brix

Refractometer



Specific Gravity

The specific gravity (SG) is the weight of the known volume divided by an equal volume of water at a specific temperature. The more dissolved sugar in the wine, the greater is the specific gravity. Winemakers commonly use a hydrometer to read in specific gravity, but these levels may be measured more accurately using a pycnometer.

Specific Gravity (SG) = $\frac{\text{wt. with sample} - \text{wt. empty pycnometer}}{\text{wt. with water} - \text{wt. empty pycnometer}}$

wt. with water – wt. empty pycnometer

Sugar Levels below 2.2° Brix

Lane and Eynon Method: A titration method used for determining sugar.

Reagents

- Fehling's Solution A: Dissolve 34.6grams of copper sulphate in distilled water and makeup to 500ml volume
- Fehling's Solution B: Dissolve 173 grams sodium potassium tartrate and 50grams sodium hydroxide pellets in about 200ml of distilled water and make up to 500ml.
- Mixed Fehling's Solution: Mix well 50ml equal volume of solution A and B.
- 1% methylene blue indicator: Dissolve 1gram of methylene blue in 100ml of distilled water.
- d -Glucose (dextrose) Solution: Dissolve 1.25 grams of d- glucose in 50 ml water, add 60ml of pure ethyl alcohol and makeup to 250ml with water. This solution will keep only for two weeks.

Note: Fehling's Solution A and B are stable for at least two months and the mixed solution for only 24hours.

Method

1. Standardise the mixed solution – Pipette 10ml of the mixed solution into a 250ml flask containing 40ml of water and a few glass beads.
2. Boil for 15 seconds and add 5 drops of 1% methyl blue indicator. Continue titrating with d-glucose solution from the burette while still boiling until the blue colour dissipates and a definite red colour persists. The number of d- glucose used in titration is the standard of the mixed Fehling's solution. Record as "Volume A"
3. Boil 20 ml of the sample for 20 seconds, cool and pipette 10ml into 250ml flask containing 40ml of water and a few glass beads. Now add 10ml of the mixed solution and bring to boil. Add a few drops of 1% methylene blue indicator and boil for 15 seconds.
4. Continue titrating with d-glucose solution from the burette until the blue colour dissipates and a definite red colour persists. The number of ml used in titration is recorded as "Volume B."
5. Calculate as: $(\text{Vol A} - \text{Vol B}) \times 0.5 = \text{Sugar content g/L (grams per litre)}$

Notes:

- Red wine samples need to be de-colourised first by adding 0.5g activated charcoal, mixing well and filtering through Whatman No1 filter paper.
- Wines with more than 10g/L of residual sugar should be diluted to one-tenth (1ml sample in 9ml water). The result from Step 5 above must then be multiplied with 10.

Clinitest Method

Clinitest kits are used by diabetics to test the sugar levels. They are convenient for low-level sugars in wines. This method gives approximate results; however, it can be used for routine sugar checks in cellars, being simple and less time-consuming. One reagent tablet is used to measure sugar concentration up to 5%. The tablets contain copper sulphate, which reacts with the sugar present in the wine.

Procedure:

1. Using an eyedropper place, 10 drops of wine sample in a small test tube

2. Drop one reagent tablet in the tube and observe the colour change, make sure not to shake the during this time.
3. The resulting colour change is compared to a colour chart provided with the tablets to determine the amount of sugar present.

Note: For wines with 1-5 % of sugar, place two drops of wine diluted with eight drops of water in the test tube. Then perform the test as directed above. If the colour change does not go past orange, wait for the reaction to finish and compare with the 2 Drop Method colour chart. The result is then multiplied by 5 to determine sugar concentration.

HOW TO DETERMINE SULPHUR DIOXIDE LEVELS

One of the most important things for the winemakers is to manage the sulphur levels to produce high-quality wines. Free sulphur oxide is the effective antioxidant and preservative. Its level is of more significance to the winemaker. Methods that can measure both free and bound sulphur dioxide are the 'Ripper Method' – adequate for routine, although it tends to give a slightly lower reading and the 'Aspiration Method' – though more complex and time-consuming, is recommended for the final check prior to bottling.

The Ripper Method

Reagents

- 25% sulphuric acid: Carefully and slowly pour 500ml of concentrated sulphuric acid in a beaker containing 1.5 litres of water. Allow it to cool before use.
- 10% sodium hydroxide: Dissolve 200g of sodium hydroxide pellets in 1.5 litres of water in a large beaker. Allow it to cool and dilute to 2 litres of distilled water.
- Starch indicator: Commercially available or boil 1g starch in 100ml water for 3 minutes. Allow it to cool before use.
- Standard 0.1N iodine solution: Dissolve 24g of potassium iodide in 150ml water and add 12.8g of iodine. Makeup to 1 litre with water. Store in a brown glass bottle in the dark. The solution is stable for 3 months.
- 0.02 iodine: Dilute 400ml of 0.1N iodine solution with 2-litre water. It is stable for four weeks in a cool dark place
- 0.02N sodium thiosulphate: Dissolve 1.24g sodium thiosulphate in 50ml of hot water. Allow it to cool and dilute to 250ml water. Add a few drops of carbon disulphide to improve stability. This solution is stable for four weeks.

Note:

- Gloves and goggles are recommended when using this acid.
- The concentrated sodium hydroxide solution should be kept in a glass jar with rubber and not glass stopper.

Method

1. Pipette 50ml of sample into a 250ml conical flask containing about 50ml water.
2. Add 10ml of 10% of sodium hydroxide solution
3. Stopper the flask and allow it to stand for 15 minutes
4. Add 10ml of 25% sulphuric acid and 2-3 ml of the starch indicator
5. Pipette with 0.02N of iodine from a burette until the solution persists to purple colour for 15-20 seconds.
6. The number of ml of iodine used in titration multiplied by 12.8 is equivalent to the ppm of total sulphur dioxide in the sample.

Notes:

- The free sulphur dioxide level is measured using the same method described above but missing out step 2 (not adding 10ml of 10% sodium hydroxide solution). It is expressed as 'ppm' free sulphur dioxide.
- The bound sulphur dioxide is the difference between the value of free and total sulphur.

Checking the normality of iron: Add 2-3 ml starch indicator to 50ml of sodium thiosulphate solution in a 250ml flask. Titrate with 0.02N iodine from a burette until the solution persists to purple colour for 15-20 seconds. Adjust to 0.02N if required with water or 0.1N iodine.

Normality = $0.02 \times 50/\text{ml of titrate}$



The Aspiration Method

Reagents

- 0.3% Hydrogen Peroxide: Pipette 10ml of 100-volume hydrogen Peroxide solution into a volumetric flask and dilute to 100ml with distilled water. Refrigerate it to prevent degradation.
- 25% Phosphoric acid: Measure 29.4ml of 85% Orthophosphoric acid to 100ml with water. Store in a sealed bottle at room temperature
- Indicator solution: Add 0.1g methyl red to 50ml of 50% aqueous ethanol. Makeup 100ml with a further 50% ethanol.
- 0.01N sodium hydroxide: Dissolve 0.4g in 500ml of water and further dilute with 1-litre water.

Method A – 'Free' Sulphur dioxide.

1. Pipette 10ml of 0.3% hydrogen peroxide into the two-neck pear-shaped flask.
2. Add 3 drops of indicator solution. If the colour is bluish to purple add 0.01N sodium hydroxide one drop at a time until the solution colour turns olive green.
3. Pipette 20ml of the sample into the round-bottomed flask and add 10ml of 25% phosphoric acid.
4. Attach this flask to the bottom of the apparatus and draw air through the sample for 10 minutes. The colour changes to red/purple, indicating that sulphur dioxide has been absorbed.
5. Remove the top, pear-shaped flask with bubble still attached and titrate with 0.01N sodium hydroxide to an olive-green endpoint.
6. Record the number of ml used in this titration.
7. Calculation: Free sulphur dioxide (ppm) = Number of ml of sodium hydroxide used X 16.

Method B – 'Bound' Sulphur dioxide.

1. After determining the free sulphur dioxide, replace the pear-shaped flask, including the contents.
2. Draw air to the apparatus and aspirate the sample for 10min while applying heat to the bottom of the flask containing the sample.
3. Remove once more the top flask, stop the heating and titrate with 0.01N sodium hydroxide from the burette.
4. Record the number of ml used in this titration.
5. To calculate the amount of sulphur dioxide multiply the number of ml used by 16. Express as 'ppm bound sulphur dioxide'.

